GuardianOps - Screenshots and Demonstration

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Demonstration walkthrough (about 3-4 minutes):

1. Start the stack and wait about 45s for the 200-sample baseline capture.
2. Click TERMINATE PROCESS - Guardian detects the crash and auto-recovers.
3. Click CORRUPT OUTPUT - a silent drift incident opens and waits for human approval (policy P-002).
4. Click AUTHORIZE ROLLBACK - the incident resolves.
5. Click REPLAY LABELLED DATASET - confusion matrix and metrics.
6. Review the auto-generated incident report.

Source and run instructions: https://github.com/kasmya/GUARDIAN-OPPS



Fig 1. Dashboard at steady state: zero incidents, confidence 100/100, live score trace tracking the baseline.



Fig 2. After a chaos-injected process crash: service DOWN, Guardian responding, crash counted.

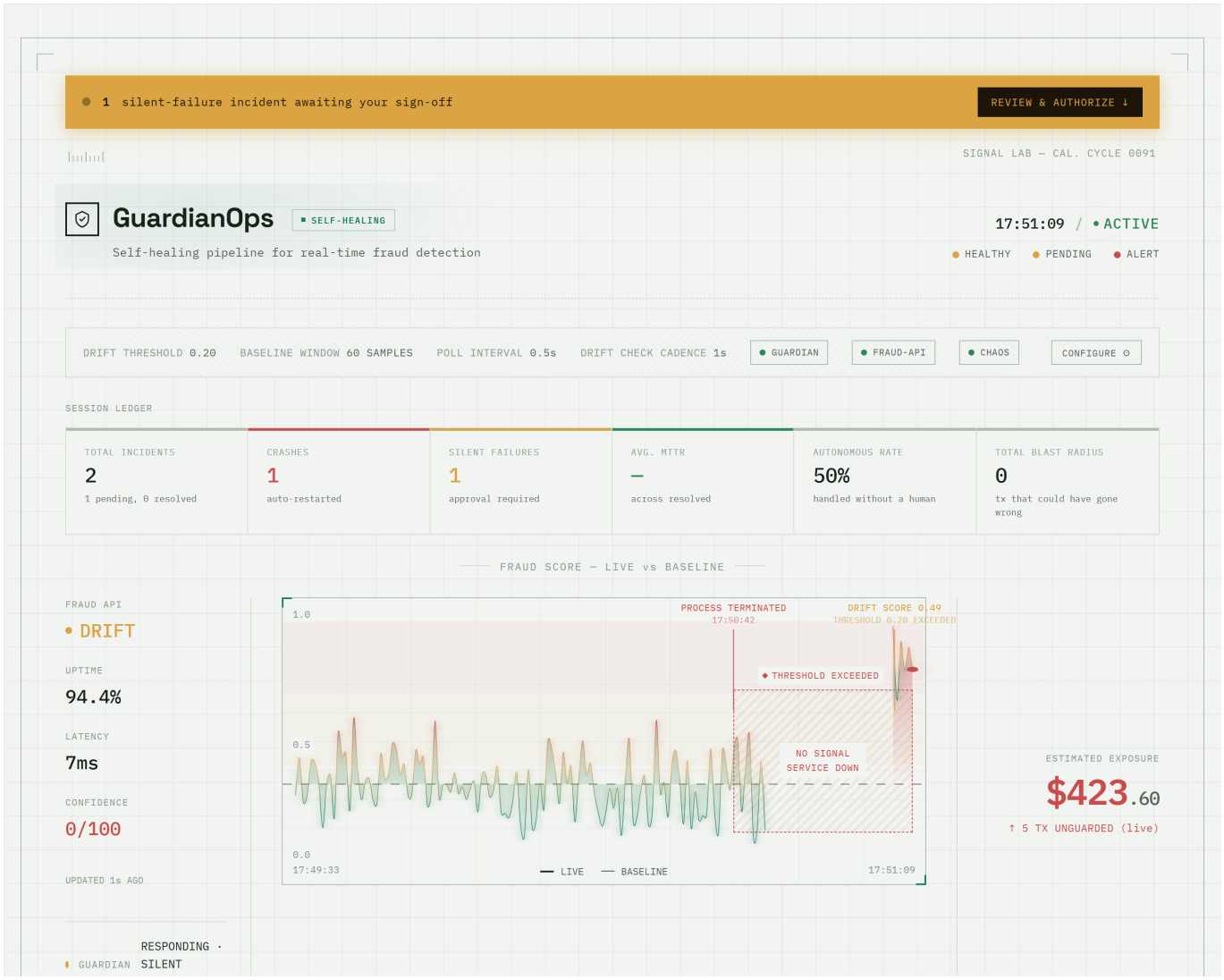


Fig 3. Silent-failure (output drift) incident awaiting human sign-off; no automatic action taken.

● PENDING AUTHORIZATION (1)

ACTING AS A. SHARMA (no auth)

Fraud API is returning scores that don't match its baseline.

Guardian classified this as a **severe distribution shift** after seeing 5 transactions with unusual scores. The service didn't crash and there's no latency spike — but the answers it's giving are different from what it gave when we know it was healthy. Rolling back returns it to that known-good behaviour.

DRIFT SEVERITY	IMPACT SO FAR	FIRST DETECTED
0.49 / 0.20	5 tx	17:51:07

− WHAT DOES "DRIFT" MEAN HERE?

Guardian keeps a snapshot of what fraud scores look like when the API is healthy — a distribution captured at startup from 200 clean transactions. Every few seconds it compares the last window of live scores against that snapshot. When the two distributions no longer match, we call it *drift*. Big drift usually means one of three things: a bad model deploy, a data-pipeline bug feeding the model wrong inputs, or the model itself getting silently corrupted. None of those crash the service, which is what makes them dangerous.

− SEE THE NUMBERS

PSI	0.495 (alert threshold 0.20; 247% of it)
KS p-value	2.70e-6
Baseline window	200 samples (frozen at startup)
Live window	60 samples (rolling)

Rule used: an incident opens only when the Kolmogorov-Smirnov test rejects "same distribution as the baseline" (`p < 0.0001`) *and* the PSI (Population Stability Index, the drift size measure bank risk teams use) is at least the threshold, on `3` consecutive checks. The KS test guards against false alarms; PSI says how large the shift is.

+ WHY DOES GUARDIAN ASK BEFORE ROLLING BACK?

+ FULL GUARDIAN REASONING (RAW)

AUTHORIZE ROLLBACK — #0002 →

APPROVAL HISTORY

No approvals recorded yet.

Fig 4. Pending Authorization card for the drift incident (policy P-002).

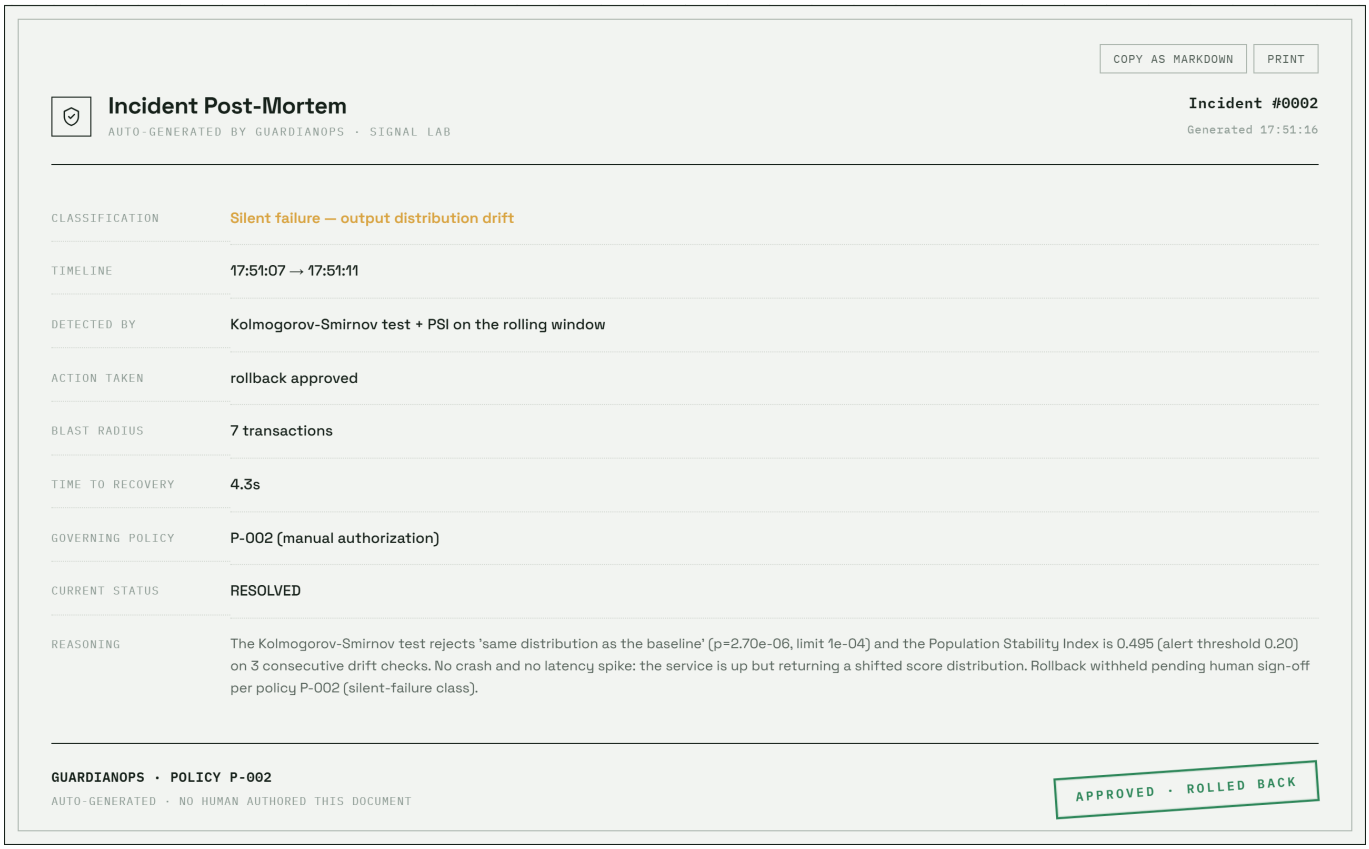


Fig 5. Auto-generated incident report: classification, timeline, MTTR, policy and reasoning.

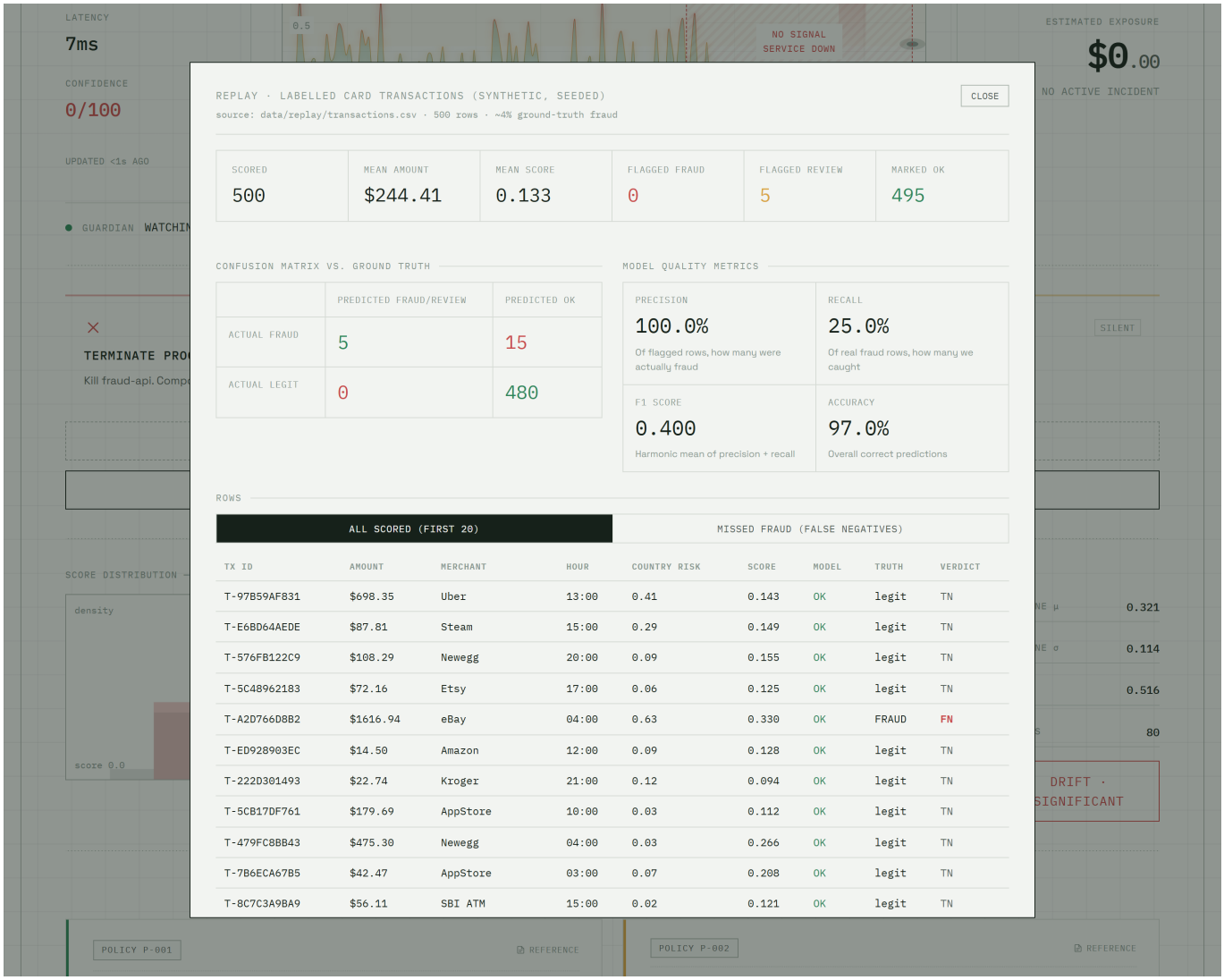


Fig 6. Replay of the labelled dataset: confusion matrix with precision, recall, F1 and accuracy.